

PROJECT DESIGN DOCUMENT

Pasco RV Resort — MLE WWTP

120,000 GPD · 2×125K gal MLE-IFAS · CRS · AI-ready

Effluent Target: "d10 mg/L NO *f*-N

Site: Pasco RV Resort · Pasco County, Florida

Document type: Preliminary / Owner Budget Package

Date: July 10, 2026

Process: MLE + Fixed PVA IFAS + CRS NBG + NH₃/NO₃/TSS + AI optimization

Prepared for: Design, permitting, vendor RFQ, and MooreVIEW SCADA integration

Estimates are planning-level (±25–35%) until site-specific engineering and vendor quotes are obtained.

1. Executive Summary

This document defines a wastewater treatment plant (WWTP) for Pasco RV Resort (Pasco County, Florida), rated for 83 GPM (120,000 GPD average flow). The biological process is Modified Ludzack-Ettinger (MLE) biological nitrogen removal configured as hybrid integrated fixed-film activated sludge (IFAS) with fixed PVA foam carriers, CRS (Chemical Reduction Systems) nanobubble generators, s::can recirculating optical measurements (BOD, NH *f*, NO *f*, TSS, pH, ORP per phase), and AI-based process optimization.

Configuration: 2×125,000 gal Integrated MLE Trains (parallel). The design target is "d10 mg/L nitrate-nitrogen (NO *f*-N) in the final effluent, with ammonia-nitrogen (NH *f*-N) <1 mg/L and secondary quality BOD .../TSS "d10 mg/L.

Metric	Value
Design average flow	120,000 GPD (83 gpm)
Design peak flow	240,000 GPD
Biological volume	250,000 gal
Footprint (approx.)	2,500-3,500 ft²
Mid-tier installed CAPEX	~\$1.44M
Total project budget (with contingency)	~\$1.75M
Annual OPEX (mid)	\$45,000-65,000

2. Design Basis

Parameter	Value	Notes
Average flow (Q)	83 GPM (120,000 GPD)	83 gpm design
Peak flow	240,000 GPD	Diurnal / seasonal peaking
Influent BOD ...	250 mg/L	Carbon for denitrification
Influent BOD ...	250 mg/L	s::can INF_BOD · C:N ratio for denitrification
Influent TSS	220 mg/L	s::can INF_TSS · AI baseline
Influent TKN	45 mg/L	RV resort strength
Influent NH <i>f</i> -N	35 mg/L	
Influent total nitrogen	50 mg/L	Includes organic N
Design minimum temperature	15°C	Florida winter design
Effluent NO <i>f</i> -N limit	"d10 mg/L	Monthly average permit target
Effluent NH <i>f</i> -N	<1 mg/L	Design objective
Effluent BOD ... / TSS	"d10 / "d10 mg/L	Secondary treatment standard

3. Process Description

Wastewater passes through preliminary screening (if required by local code), then enters the pre-anoxic zone of the MLE reactor. Return activated sludge (RAS) is mixed with influent in the anoxic zone where denitrification occurs using influent readily biodegradable COD (rbCOD). Mixed liquor underflows to the aerobic IFAS zone where autotrophic nitrifiers oxidize ammonia on both suspended floc and fixed PVA biofilm. Internal recycle (IR) pumps return nitrified mixed liquor from the aerobic zone to the anoxic zone.

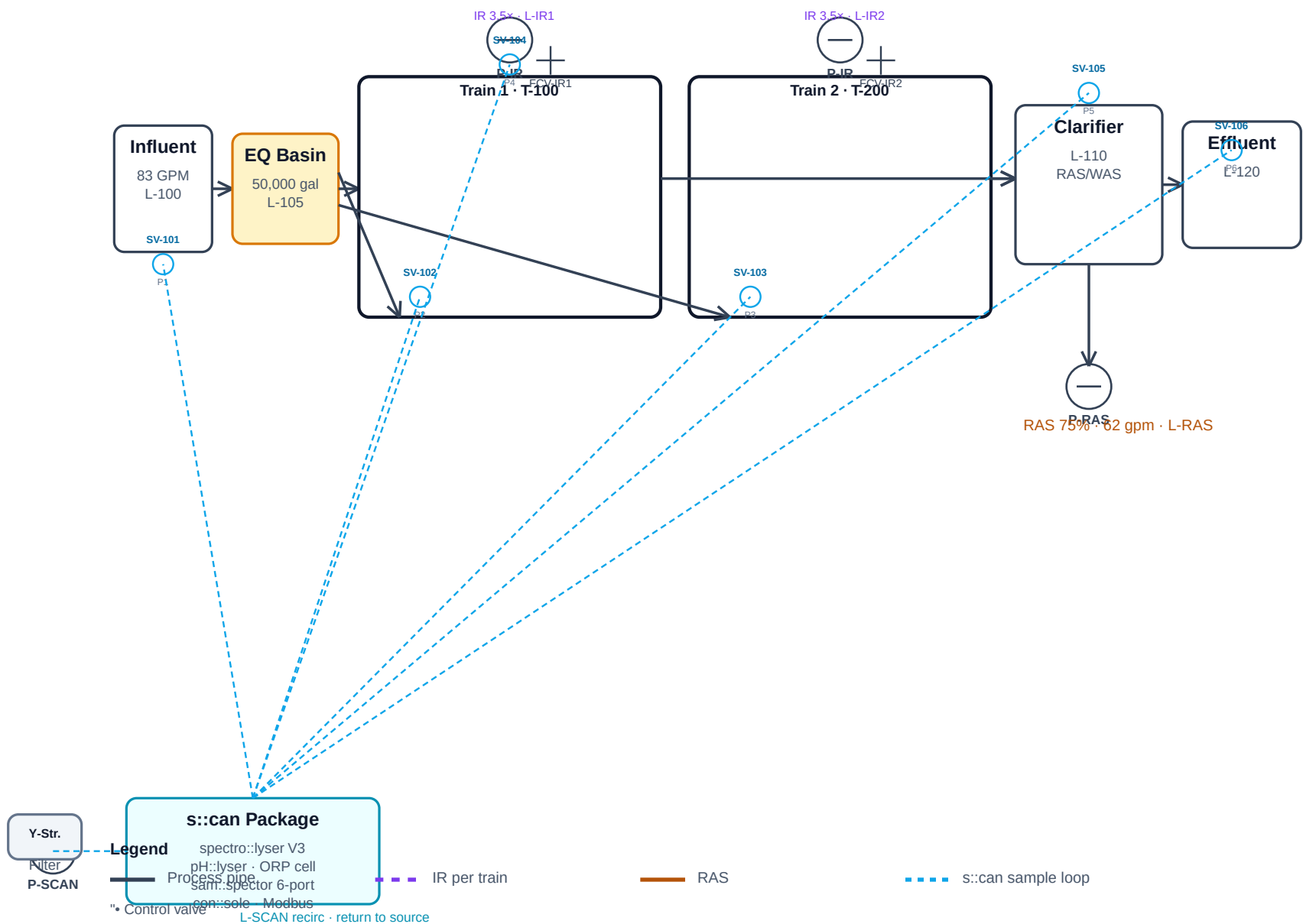
Oxygen is supplied primarily through a continuous sidestream loop: mixed liquor is pumped through a CRS Nanobubble Generator (NBG) and returned to the aerobic zone. A small fine-bubble diffuser and trim blower maintains dissolved oxygen (DO) setpoint during peak load or cold weather. Nano-bubbles are not applied in the anoxic zone.

Online measurements use a recirculating scan optical analyzer package (spectrolyser, pHlyser, ORP in flow cell, sam: spector sequencer) that cycles sample ports at each MLE phase: influent (BOD, COD, TSS, NH_4 , pH, UV254), pre-anoxic (NO_x , ORP, pH, TSS, residual COD), IR stream, aerobic outlet, and final effluent. Influent BOD enables carbon-limited denitrification detection alongside NH_4 / NO_x /TSS. Aerobic DO remains on dedicated continuous probes for blower PID. All phase tags historize in MooreVIEW; effluent NO_x -N drives IR VFD trim; AI optimizer uses INF_BOD, C:N ratio, and phase pH/ORP within operator bounds.

Treated mixed liquor flows to an integrated lamella/tube settler. Clarified effluent is discharged; RAS returns to anoxic; waste activated sludge (WAS) is removed on an SRT schedule.

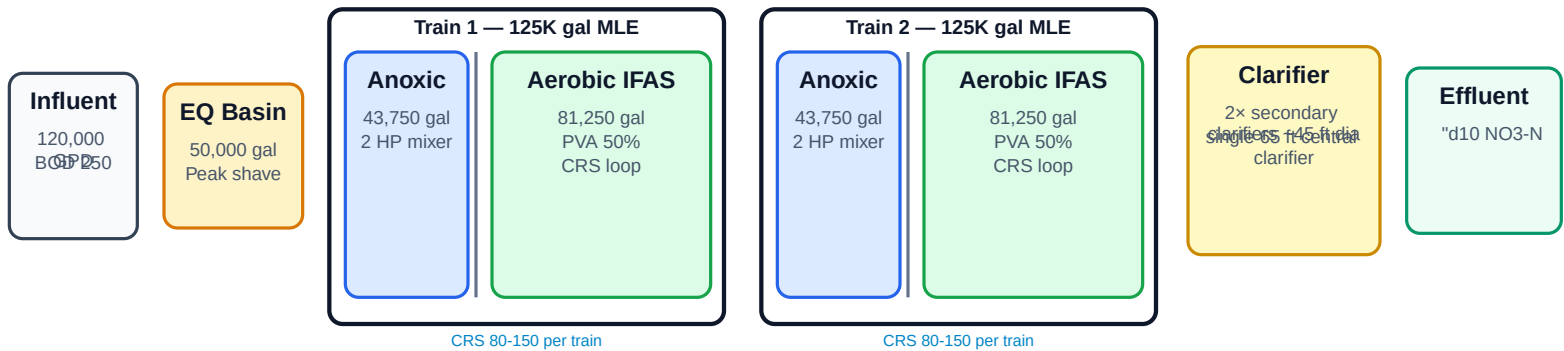
4. Piping & Instrumentation Diagram (P&ID)

83 GPM · Piping, valves, pumps, recycle loops



5. Mechanical Drawing

Equipment layout, volumes, materials · 2×125,000 gal Integrated MLE Trains (parallel)



s::can Package

Legend

spectro::lyser V3

pH::lyser · ORP cell

2× parallel 125K gal MLE trains

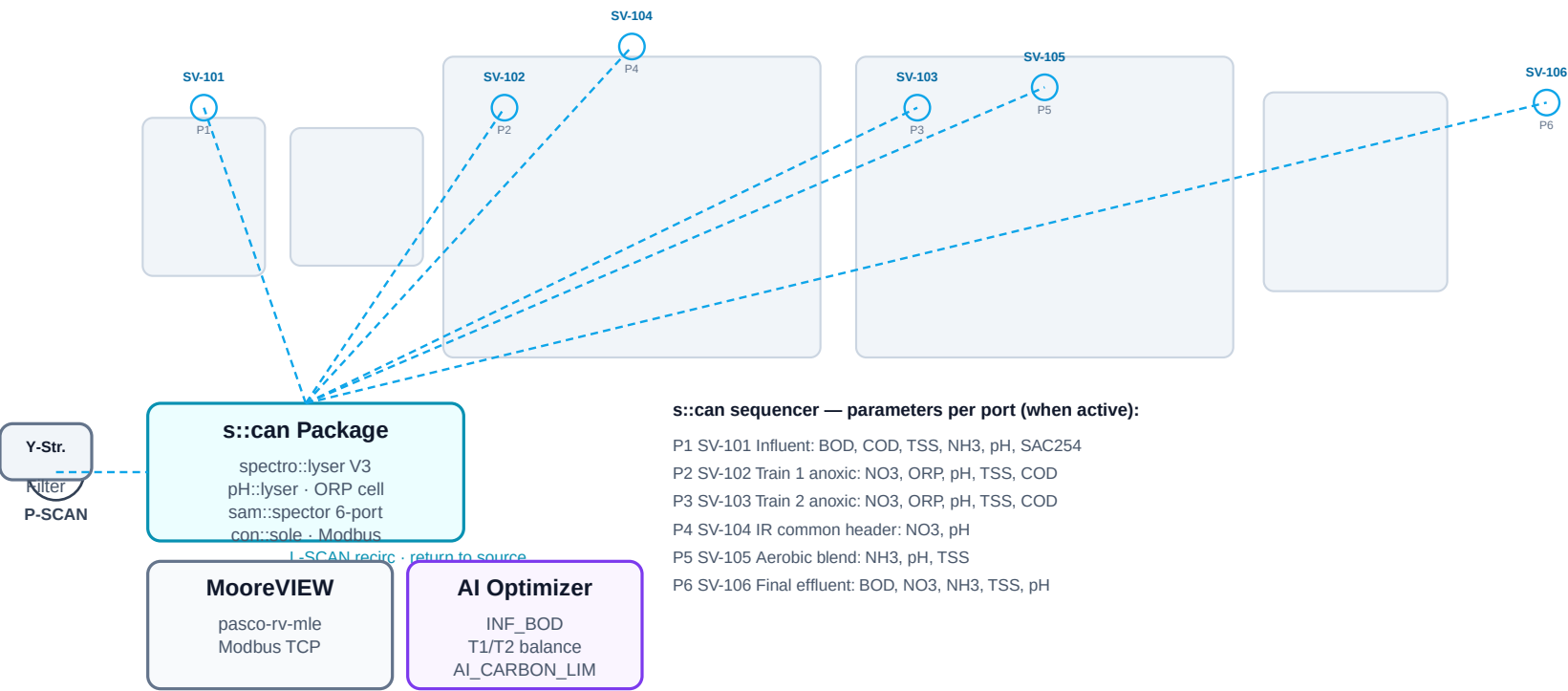
sam::spectro 6-port

con::sole · Modbus

ent ~2,500-3,500 ft² · Blower 7.5 HP
EQ basin for seasonal peaking
Shared clarifier + effluent

6. Instrumentation Diagram

s::can recirculating optical loop · per-phase BOD/pH/ORP · MooreVIEW pasco-rv-mle



7a. Process Flow Summary

Refer to the drawing set on the preceding landscape pages: Sheet 1 (P&ID) includes the s::can recirculating sample loop (L-SCAN, P-SCAN, SV-101...106) to each phase; Sheet 2 (Mechanical) shows the analyzer shelter; Sheet 3 (Instrumentation) lists optical parameters per sequencer port. Main process: influent enters pre-anoxic with RAS; denitrified mixed liquor passes to aerobic IFAS; nitrified liquor is settled and discharged. IR (3.5× Q) returns nitrate-rich liquor to anoxic. CRS sidestream provides nanobubble oxygen in the aerobic zone only.

7b. Process Measurement Points (s::can Recirculating Loop)

Tag	Sample Port	Parameter	Target / Alarm	Control Use
INF_BOD	Influent (SV-101)	BOD	Track ~250 mg/L · C:N	Plant influent after EQ
INF_COD	Influent (SV-101)	COD	Historian + AI	Plant influent after EQ
INF_TSS	Influent (SV-101)	TSS	Track ~220 mg/L	Plant influent after EQ
INF_NH3	Influent (SV-101)	NH3	Track ~35 mg/L	Plant influent after EQ
INF_PH	Influent (SV-101)	pH	6.5–8.5	Plant influent after EQ
INF_SAC254	Influent (SV-101)	SAC254	UV254 organic load	Plant influent after EQ
T1_ANOX_NO3	Train 1 anoxic (SV-102)	NO3	<5 mg/L ideal	Train 1 denitrification
T1_ANOX_ORP	Train 1 anoxic (SV-102)	ORP	-50 to +50 mV anoxic	Train 1 denitrification
T1_ANOX_PH	Train 1 anoxic (SV-102)	pH	6.5–8.5	Train 1 denitrification
T1_ANOX_TSS	Train 1 anoxic (SV-102)	TSS	2,500-4,500 mg/L	Train 1 denitrification
T1_ANOX_COD	Train 1 anoxic (SV-102)	COD	Residual carbon proxy	Train 1 denitrification
T2_ANOX_NO3	Train 2 anoxic (SV-103)	NO3	<5 mg/L ideal	Train 2 denitrification
T2_ANOX_ORP	Train 2 anoxic (SV-103)	ORP	-50 to +50 mV anoxic	Train 2 denitrification
T2_ANOX_PH	Train 2 anoxic (SV-103)	pH	6.5–8.5	Train 2 denitrification
T2_ANOX_TSS	Train 2 anoxic (SV-103)	TSS	2,500-4,500 mg/L	Train 2 denitrification
T2_ANOX_COD	Train 2 anoxic (SV-103)	COD	Residual carbon proxy	Train 2 denitrification
IR_NO3	IR common header (SV-104)	NO3	15–30 mg/L typical	Shared IR sample per operating train
IR_PH	IR common header (SV-104)	pH	6.5–8.5	Shared IR sample per operating train
AER_NH3	Aerobic blend (SV-105)	NH3	<1 mg/L	Combined aerobic outlet · AER_DO per train in basin
AER_PH	Aerobic blend (SV-105)	pH	6.5–8.5	Combined aerobic outlet · AER_DO per train in basin
AER_TSS	Aerobic blend (SV-105)	TSS	Historian + AI	Combined aerobic outlet · AER_DO per train in basin
EFF_BOD	Final effluent (SV-106)	BOD	"d10 mg/L permit	FDEP permit point
EFF_NO3	Final effluent (SV-106)	NO3	"d10 mg/L permit · IR PID	FDEP permit point

EFF_NH3	Final effluent (SV-106)	NH3	<1 mg/L	FDEP permit point
EFF_TSS	Final effluent (SV-106)	TSS	Historian + AI	FDEP permit point
EFF_PH	Final effluent (SV-106)	pH	6.5–8.5	FDEP permit point
AER_DO	Aerobic basin (dedicated)	DO	2.0 mg/L setpoint	Not on scan loop — continuous probe

s::can package: spectro::lyser V3 (UV-Vis BOD, COD, NO₃ f, NH₃ f, TSS, SAC254), pH::lyser in flow cell, ORP at anoxic-phase dwells, con::sole controller, sam::spector 6-port sequencer, P-SCAN recirculation pump with Y-strainer and 10 L buffer. Sample return to source. Dwell 10 min/port; full cycle ~60 min. Calibrate BOD/COD vs lab grab monthly; derive INF_BOD_TKN in MooreVIEW ST. Modbus TCP to edge historian at 1-min scan during active port.

7c. s::can Sequencer Port Schedule

Port	Valve	Phase	Optical Parameters	Notes
1	SV-101	Influent	BOD, COD, TSS, NH3, pH, SAC254	Plant influent after EQ
2	SV-102	Train 1 anoxic	NO3, ORP, pH, TSS, COD	Train 1 denitrification
3	SV-103	Train 2 anoxic	NO3, ORP, pH, TSS, COD	Train 2 denitrification
4	SV-104	IR common header	NO3, pH	Shared IR sample per operating train
5	SV-105	Aerobic blend	NH3, pH, TSS	Combined aerobic outlet · AER_DO per train in basin
6	SV-106	Final effluent	BOD, NO3, NH3, TSS, pH	FDEP permit point

7d. AI-Based Process Optimization

The s::can multivariate data stream enables supervised ML that predicts optimal IR ratio, WAS rate, and DO setpoint. INF_BOD and INF_COD quantify carbon available for denitrification; when EFF_NO3 is high but INF_BOD/TKN is low, the model sets AI_CARBON_LIM rather than increasing IR. Phase pH and anoxic ORP provide redox context at each MLE step.

AI Input Tag / Feature	Role in Model
INF_BOD	Influent carbon load; C:N vs TKN; carbon-limit alarm
INF_COD	rbCOD proxy; supplemental carbon decision
INF_BOD_TKN	Derived ratio — denitrification potential
INF_TSS	Solids shock; normalize MLSS
INF_NH3	Nitrogen load forecast
INF_PH	pH inhibition / nitrifier stress
ANOX_NO3 + ANOX_ORP	Denitrification state (single train)
T1_ANOX_* / T2_ANOX_*	Per-train denitrification (dual layout)
ANOX_COD	Residual carbon in anoxic — carbon limited?

IR_NO3	IR effectiveness
AER_NH3 + AER_DO	Nitrification health
EFF_NO3 + EFF_NH3 + EFF_BOD	Permit feedback · model reward
Phase pH (all ports)	Process stability across MLE phases

Recommended AI outputs (constrained setpoints written to MooreVIEW tags):

- AI_IR_SP — suggested IR ratio (bounded 2.5–4.5× Q); suppressed when AI_CARBON_LIM = 1.
- AI_WAS_SP — suggested WAS mL/min from ANOX_TSS trend vs target MLSS band.
- AI_DO_SP — suggested aerobic DO mg/L when AER_NH3 rising and ANOX_TSS stable.
- AI_CARBON_LIM — flag when INF_BOD/TKN < 3.5 or ANOX_COD residual high with ANOX_NO3 > 8 mg/L.
- AI_ALARM — anomaly score when INF_TSS or phase pH deviates >2σ from baseline.

Deployment path: (1) collect 90 days historian data with rule-based IR trim; (2) train regression/gradient model offline on NH₄-N/NO₃-N/TSS features; (3) run inference on edge PC alongside MooreVIEW ST program; (4) validate AI vs manual operation for 30 days before permit reliance. AI never bypasses hard limits on EFF_NO3 >10 mg/L or EFF_NH3 >2 mg/L — those trigger immediate rule-based IR increase and operator alert.

8. Reactor Sizing

Zone	Volume	% of Total	Function
Train 1 — Pre-anoxic	43,750 gal	35%	Denitrification per train
Train 1 — Aerobic IFAS	81,250 gal	65%	Nitrification + PVA + CRS
Train 2 — Pre-anoxic	43,750 gal	35%	Parallel redundant train
Train 2 — Aerobic IFAS	81,250 gal	65%	Parallel redundant train
EQ basin (recommended)	50,000 gal	—	Seasonal / diurnal flow equalization
Total biological	250,000 gal	100%	HRT "H 2.08 days @ avg flow

PVA foam media (fixed in SS cages, 5 mm max slot) — total fill ~10,200 ft³:

- Fill fraction: 50% of aerobic volume (~81,250 gal displaced)
- Media type: PVA foam cubes/blocks, 10–20 mm
- Surface area: 400–600 m²/m³ (vendor-specific)
- Effective biofilm SRT: 30–60+ days on carriers

9. CRS Nanobubble Aeration System

Equipment: CRS Nanobubble Generator (NBG) by Chemical Reduction Systems (chemicalreduction.com). The CRS NBG is a passive inline/sidestream device using hydrodynamic cavitation and ionization of entrained gases. It requires no electricity, no external gas input, and has no moving parts. Housing: 316 or 304 stainless steel; core alloy suitable for NSF/ANSI 61 applications. Factory warranty: 5 years; design service life: 10+ years.

Component	Specification
CRS NBG model	2" line, sidestream per train

Sidestream flow	80-150 per train continuous
Sidestream pump	3 HP, 24/7 duty
Aerobic DO setpoint	2.0 mg/L
Backup aeration	7.5 HP fine-bubble (DO trim)
Anoxic mixing	2 HP submersible mixer per train
Estimated O ₂ demand	350-450 lb O ₂ /d

10. Recycle & Solids Management

Stream	Rate	Equipment
Internal recycle (IR)	3.5× Q = 420,000 GPD (~291 gpm)	VFD centrifugal per train
RAS	75% Q = 90,000 GPD (~62 gpm)	Return to anoxic
WAS	300-600 gal/d	Timed valve; SRT 12–20 d
MLSS target	2,500-4,000 mg/L	Suspended + biofilm biomass

IR ratio is trimmed on effluent NO₃-N: increase IR when NO₃-N >10 mg/L (max 4.5×); decrease when NO₃-N <6 mg/L (min 2.5×).

11. Expected Performance

Parameter	Expected Effluent
BOD ...	5–8 mg/L
TSS	5–10 mg/L
NH ₃ -N	<1 mg/L
NO ₃ -N	6–10 mg/L (with IR trim)
Total nitrogen (TN)	10–14 mg/L

Nitrogen balance: ~35–40 mg/L NO₃-N formed after full nitrification must be reduced ~75–80% via MLE denitrification to achieve <10 mg/L NO₃-N. At BOD 200 mg/L and TKN 40 mg/L, influent rbCOD (~100–120 mg/L) is adequate without supplemental carbon under normal conditions.

12. Equipment Schedule

Item	Qty	Size / Model
Integrated MLE tank (125K gal)	2	Parallel trains; 35/65 anoxic/aerobic
EQ basin	1	50,000 gal FRP/concrete
PVA foam + retention cages	2 lots	~5,100 ft ³ per train
CRS NBG sidestream unit	2	2" SS per train; RFQ from CRS
Sidestream recirculation pump	2	80-150 per train; 3 HP
Centrifugal blower farm	1 lot	3× 7.5 HP duty/standby

Anoxic submersible mixer	2	2 HP per train
IR pump (VFD)	2	~146 gpm per train
RAS pump system	2	~31 gpm per train
2× secondary clarifiers ~45 ft dia OR single 65 ft central clarifier	1–2	Per hydraulic load
s::can recirculating analyzer package	1	s::can spectro::lyser V3 (UV-Vis) — BOD, COD, NO3, NH3, TSS, SAC254; s::can pH::lyser + ORP in flow cell (anoxic ports); s::can con::sole controller + sam::spector 6-port sequencer; Recirculating loop: P-SCAN · Y-strainer · 10L buffer · return to source; 6 sample taps (¼" SS) · 10 min dwell · full cycle ~60 min; Weather-rated analyzer shelter · Modbus TCP to MooreVIEW
MooreVIEW edge SCADA + AI module	1	Project: pasco-rv-mle

s::can installed budget ~\$85k (included in Section 16 CAPEX). Monthly lab verification of spectro::lyser BOD/ COD vs Standard Methods required for permit-grade reporting.

13. Instrumentation & Control

Tag	Sample Port	Function
INF_BOD	Influent (SV-101)	Plant influent after EQ
INF_COD	Influent (SV-101)	Plant influent after EQ
INF_TSS	Influent (SV-101)	Plant influent after EQ
INF_NH3	Influent (SV-101)	Plant influent after EQ
INF_PH	Influent (SV-101)	Plant influent after EQ
INF_SAC254	Influent (SV-101)	Plant influent after EQ
T1_ANOX_NO3	Train 1 anoxic (SV-102)	Train 1 denitrification
T1_ANOX_ORP	Train 1 anoxic (SV-102)	Train 1 denitrification
T1_ANOX_PH	Train 1 anoxic (SV-102)	Train 1 denitrification
T1_ANOX_TSS	Train 1 anoxic (SV-102)	Train 1 denitrification
T1_ANOX_COD	Train 1 anoxic (SV-102)	Train 1 denitrification
T2_ANOX_NO3	Train 2 anoxic (SV-103)	Train 2 denitrification
T2_ANOX_ORP	Train 2 anoxic (SV-103)	Train 2 denitrification
T2_ANOX_PH	Train 2 anoxic (SV-103)	Train 2 denitrification
T2_ANOX_TSS	Train 2 anoxic (SV-103)	Train 2 denitrification
T2_ANOX_COD	Train 2 anoxic (SV-103)	Train 2 denitrification
IR_NO3	IR common header (SV-104)	Shared IR sample per operating train
IR_PH	IR common header (SV-104)	Shared IR sample per operating train
AER_NH3	Aerobic blend (SV-105)	Combined aerobic outlet · AER_DO per train in basin

AER_PH	Aerobic blend (SV-105)	Combined aerobic outlet · AER_DO per train in basin
AER_TSS	Aerobic blend (SV-105)	Combined aerobic outlet · AER_DO per train in basin
EFF_BOD	Final effluent (SV-106)	FDEP permit point
EFF_NO3	Final effluent (SV-106)	FDEP permit point
EFF_NH3	Final effluent (SV-106)	FDEP permit point
EFF_TSS	Final effluent (SV-106)	FDEP permit point
EFF_PH	Final effluent (SV-106)	FDEP permit point
AER_DO	Aerobic basin (dedicated)	Not on scan loop — continuous probe
SCAN_ACTIVE_PORT	Sequencer	Current port 1–N
SCAN_RUN / SCAN_FAULT	P-SCAN loop	Pump status / system fault
AER_DO	Aerobic basin	Continuous DO 2.0 mg/L; blower PID
AI_IR_SP	AI Optimizer out	Suggested IR setpoint (Auto mode)
AI_WAS_SP	AI Optimizer out	Suggested WAS rate from TSS trend
AI_CARBON_LIM	AI Optimizer out	Carbon-limited denitrification flag
IR_FLOW / IR_RATIO	IR pump	Flow and ratio monitoring
TANK_LVL	Reactor	Level alarm
NANO_LOOP_RUN	Sidestream pump	CRS loop runtime / fault

Control logic summary:

- Sidestream CRS loop: run continuous (24/7).
- s::can sequencer: run continuous; alarm SCAN_FAULT; log all ports each cycle.
- IF EFF_NO3 > 10 mg/L AND NOT AI_CARBON_LIM !' increase IR +0.2×/hr (max 4.5×).
- IF AI_CARBON_LIM = 1 !' hold IR; alarm operator; evaluate supplemental carbon.
- IF EFF_NO3 < 6 mg/L !' decrease IR " 0.2×/hr (min 2.5×).
- IF AER_NH3 > 2 mg/L OR EFF_NH3 > 1 mg/L !' increase DO or reduce WAS.
- IF ANOX_TSS > 4,000 mg/L for 4 hr !' increase WAS per AI_WAS_SP; check settler.
- IF INF_BOD/TKN < 3.5 (derived tag) !' set AI_CARBON_LIM; do not raise IR.
- IF ANOX_ORP > +100 mV at anoxic port dwell !' alarm: denitrification lost.
- IF phase pH < 6.2 or > 8.8 at any port !' process pH alarm.

14. Tank Layout (Section)

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+--- ANOXIC 87,500 gal -----+--- AEROBIC 162,500 gal + PVA ---+
| Influent + RAS -> Mixer | Fixed PVA foam cages (50% fill)|
| <- IR from aerobic     | CRS sidestream + backup diff.  |
| ORP probe              | DO probe                       |
+-----+-----+-----> Settler -> Eff

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15. Startup & Commissioning

- Fill reactor; seed with nitrifying return sludge or commercial culture.
- Install and cage PVA media permanently — floating media causes clarifier carryover.
- Commission anoxic mixer and IR/RAS loops before enabling CRS sidestream.
- Start CRS sidestream at 50% pump speed; ramp while monitoring AER_DO and ANOX_ORP.
- Allow 2–3 weeks biofilm growth on PVA before enforcing 10 mg/L NO_x-N limit.
- Tune IR trim after 30 days of stable operation and seasonal temperature data.

16. Capital Cost Estimate (CAPEX)

Planning-level installed costs, USD, 2025–2026. ±25–35% until vendor quotes received.

Category	Budget	Mid	High
MLE reactor tanks (2×125K or 250,000 gal)	\$187k	\$316k	\$556k
Fixed PVA foam + cages	\$47k	\$115k	\$222k
CRS sidestream aeration (all trains)	\$56k	\$100k	\$200k
Pumps, mixers, clarifiers	\$112k	\$215k	\$400k
Instrumentation + AI	\$75k	\$172k	\$334k
Controls / SCADA	\$47k	\$86k	\$178k
Electrical + piping + site	\$140k	\$258k	\$489k
Install + commissioning	\$93k	\$172k	\$311k
Engineering + FDEP permits	\$75k	\$144k	\$267k
TOTAL INSTALLED	\$933k	\$1435k	\$2224k

17. Project Budget Summary

Line Item	Amount
Construction (mid installed)	~\$1.44M
Owner contingency (15%)	~\$215,250
Collection / EQ / site civil (allowance)	~\$215,250
First-year OPEX reserve	~\$28,700
TOTAL PROJECT BUDGET	~\$1.75M

Excluded from above: land acquisition, long collection systems, enclosed buildings, effluent UV/chlorination, and septage receiving.

18. Operating Cost Estimate (OPEX)

Item	Annual Cost
Power (CRS loop + trim blower)	\$120 – \$180

Lab / compliance sampling	\$800 – \$2,500
Analyzer consumables	\$0 – \$1,500
WAS hauling	\$200 – \$600
PVA media replacement reserve	\$300 – \$800
Maintenance contract	\$500 – \$2,000
Contract O&M visits	\$1,000 – \$3,000
TOTAL ANNUAL OPEX	\$45,000-65,000

19. CRS Vendor RFQ Template

To: Chemical Reduction Systems (CRS) – chemicalreduction.com
Subject: RFQ – CRS NBG for 120,000 GPD MLE-IFAS WWTP (Pasco RV Resort)

Application: 120,000 GPD MLE-IFAS, 2x125,000 gal Integrated MLE Trains (parallel), fixed PVA biofilm carriers, effluent target <=10 mg/L NO3-N.

Request: 2x CRS NBG sidestream unit(s), 2" line, 316 SS, 80-150 per train continuous recirculation on mixed liquor per train.

Please quote: unit price, min/max flow, pressure drop, lead time, warranty, WWTP reference installations, and recommendation for backup aeration at 10°C.

20. Permitting & Design Notes

- Pasco County / FDEP: confirm NPDES or individual wastewater permit for RV resort classification.
- Design for seasonal peaking (snowbird occupancy); EQ basin recommended for diurnal/weekly surges.
- RV influent: elevated grease and surfactants — bar screen + grease interceptor upstream; monitor ANOX_TSS.
- Florida min temp 15°C supports higher nitrifier rates; size SRT for 12-day winter equivalent.
- Pasco County reuse/disposal: confirm effluent disposal (surface water, reuse, or deep well).

21. Document Revision

Rev	Date	Description
0	2026-07-10	Initial preliminary design package